

Lorenzo Bianchi

PHD STUDENT IN VISION-LANGUAGE ARTIFICIAL INTELLIGENCE

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I am a passionate Ph.D. candidate in Vision-Language Artificial Intelligence at the University of Pisa and an Associate Researcher at ISTI-CNR. My work has been published at top-tier venues such as CVPR (three publications with one as a Highlight Poster), ICCV, and WACV, and I have received Best Paper Awards at CBMI and IEEE-CH. In 2025, I joined Disney Research | Studios in Zürich as a Research Intern, where I worked on large-scale diffusion models for text-conditioned image generation. My research interests focus on Vision-Language Models, particularly in understanding, improving, and exploiting the representations learned by foundational models such as CLIP and DINO for image-text tasks.

Education

Ph.D. in Vision-Language Artificial Intelligence

UNIVERSITY OF PISA

Pisa, Italy

Mar. 2023 – Present

- Research interests: Vision-Language AI, Representation Learning, and Unsupervised Learning.
- Supervisors: Dr. Fabrizio Falchi, Dr. Fabio Carrara, Dr. Nicola Messina, Dr. Giuseppe Amato.

M.Sc. in Artificial Intelligence and Data Engineering

UNIVERSITY OF PISA

Pisa, Italy

Sep. 2020 – Feb. 2023

- Final Mark: **110/110**.
- Thesis Title: "Design and development of cross-modal retrieval techniques based on transformer architectures".

B.Sc. in Computer Engineering

UNIVERSITY OF PISA

Pisa, Italy

Sep. 2017 – Sep. 2020

- Final Mark: **110/110**.
- Thesis Title: "Dynamics analysis of sports events on online social networks through clusters".

Work Experience

Associate Researcher

CNR (NATIONAL RESEARCH COUNCIL OF ITALY)

Pisa, Italy

Mar. 2023 – Present

- Research Associate in deep learning for vision-language tasks at the Artificial Intelligence for Media and Humanities (AIMH) Lab within the Institute of Information Science and Technologies "Alessandro Faedo" (ISTI).
- Focused on studying, enhancing, and exploiting foundational model representations (CLIP, DINO, etc.) and applying them to a variety of tasks.
- Worked on image-text matching, retrieval, open-vocabulary object detection, open-vocabulary semantic segmentation, and image captioning.
- Collaborated in the Sun XR European project on extended reality and in the ITSERR project, advancing AI applications in Religious Studies.

Research Intern

DISNEY RESEARCH | STUDIOS

Zürich, Switzerland

May 2025 – Aug. 2025

- Designed, implemented, and trained large-scale diffusion models for image generation.
- Supervised by Dr. Vinicius C. Azevedo.

Machine Learning Engineer

WEVO S.R.L.

Pisa, Italy

Apr. 2024 – Jul. 2024

- Developed a sentiment analysis system for social media posts, implementing deep learning models for automatic content classification.

Publications

One Patch to Caption Them All: A Unified Zero-Shot Captioning Framework

CVPR 2026

Denver, Colorado

L. Bianchi*, G. Pacini*, F. Carrara, N. Messina, G. Amato, F. Falchi
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Reviving ConvNeXt for Efficient Convolutional Diffusion Models

CVPR 2026

Denver, Colorado

T. Kwon, L. Bianchi, L. Wittke, F. Watine, F. Carrara, J.C. Ye, R.M. Weber and V.C. Azevedo
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

CountingDINO: A Training-free Pipeline for Class-Agnostic Counting using Unsupervised Backbones

WACV 2026

Tucson, Arizona

G. Pacini*, L. Bianchi*, L. Ciampi, N. Messina, G. Amato, F. Falchi
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

Talking to DINO: Bridging Self-Supervised Vision Backbones with Language for Open-Vocabulary Segmentation

ICCV 2025

Honolulu, Hawaii

L. Barsellotti*, L. Bianchi*, N. Messina, F. Carrara, M. Cornia, L. Baraldi, F. Falchi, R. Cucchiara
IEEE/CVF International Conference on Computer Vision (ICCV)

- **ReCoptic: Computer Vision for the Reconstruction of Dismembered Coptic Codices**
L. Bianchi, A. Moreo, F. Falchi, F. Sebastiani, C. Bianchi
IEEE International Conference on Cyber Humanities (IEEE CH) (Best Paper Award) IEEE CH 2025
Florence, Italy
- **The Devil is in the Fine-Grained Details: Evaluating Open-Vocabulary Object Detectors for Fine-Grained Understanding**
L. Bianchi, F. Carrara, N. Messina, C. Gennaro, F. Falchi
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (Highlight Paper) CVPR 2024
Seattle, Washington
- **Is CLIP the main roadblock for fine-grained open-world perception?**
L. Bianchi, F. Carrara, N. Messina, F. Falchi
International Conference on Content-Based Multimedia Indexing (CBMI) (Best Paper Award) CBMI 2024
Reykjavík, Iceland

Awards

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|------|---|--------------|
| 2026 | Outstanding Reviewer , Recognized for exceptional contributions as a reviewer at CVPR | CVPR |
| 2025 | Best Paper Award , ReCoptic: Computer Vision for the Reconstruction of Dismembered Coptic Codices | IEEE CH |
| 2024 | Best Paper Award , Is CLIP the main roadblock for fine-grained open-world perception? | CBMI |
| 2024 | Highlight Paper , The Devil is in the Fine-Grained Details: Evaluating Open-Vocabulary Object Detectors for Fine-Grained Understanding | CVPR |
| 2015 | Team Winner , ZeroRobotics Italian High School Tournament 2015 (Teamleader of TeamOre) | Turin, Italy |

Activities

SUMMER SCHOOLS

International Computer Vision Summer School

ATTENDEE

Sicily
July 2024

ELLIS Summer School on Large-Scale AI for Research and Industry

ATTENDEE

Modena
September 2023

PEER REVIEWING

Conferences

ICMR (2023, 2025), BMVC (2025), CVPR (2026 - Outstanding Reviewer), ECCV (2026), NeurIPS (2026), IJCV (2026)

2023 - Present

HACKATON

CTF Competitions

Member of a CTF Time ranked team (Yupsilon team).

2021-2022